

Modern Computing Techniques - S2, 2025

Presentation and Final Project

This document contains the instruction for assessment 2 (presentation) and 3 (final report).

For these assessments, you need to select a scientific article and implement some of the methods you have learned in MXN442 and compare it with the article.

You will present your work either in week 12 or 13 during the lecture time (we will set the schedule closer to the time) and submit a final report. You must submit the presentation and final report on Canvas no later than **October 17th, 23:59pm**.

Paper Selection:

- **Objective:** Students select a peer-reviewed article that uses data science or statistical machine learning algorithms to address a problem of interest.
- **Criteria:** You can find articles on [Google Scholar](#) or [Scopus](#). The article should ideally be published in reputable journals or conferences (e.g., NeurIPS, ICML, Machine Learning, etc.). Check the journal quality via [Scimago Journal Rank](#) to ensure it's from a Q1 journal.

Articles must:

- Be published **after 2020**
- Be relevant to concepts learned in MXN442 (regression or classification broadly)
- Have **data available**
Choose an article that's neither too simple nor overly complex, with datasets of a **decent size** (hundreds of observations or more).
- **Feedback:** Students are welcome to seek feedback on their chosen article to ensure it meets the requirements.

Outline of the project and presentation

Your project report must be **less than 10 pages** long. It should include:

1. **Front page:** Title of the article you have chosen (provide the full reference and link to the article at the end of the report), your name, student ID.
2. **Introduction:** Brief description of the problem background, the main research questions or objectives of the study, novelty.
3. **Data:** Description of data, including any descriptive analytics and graphs appropriate.

4. **Methodology:** Brief description of the solution technique used in the paper, including key equations. Then, implement **three methods** from discussed methods in MXN442, including:
 - Preprocessing
 - Model Implementation
 - Optimisation and cross-validation
 - Evaluation
5. **Results:** Report your results and whether your results are comparable with the paper or not. Use appropriate tables and graphs. You do not have to get the same results as you may be implementing a simple or more complicated model.
6. **Discussion:** Include what ideas you have for improving the solution techniques and implement only one. For this section you can use GenAI.
 - Use GenAI to brainstorm possible extensions or improvements to the paper (e.g., alternative models that maybe a better fit, metrics, or architectures).
 - Select one idea and use GenAI to implement it fully as part of your project.
 - Keep your GenAI prompts and outputs and include an analysis on:
 - Where GenAI's suggestions worked well.
 - Where GenAI failed, hallucinated, or gave unusable suggestions/results.
 - What did you think that GenAI did not think of, and vice versa?
 - If you have used GenAI for section 4, then report it here that you have used it .
 - You can find a guide for prompt engineering here:
<https://platform.openai.com/docs/guides/prompt-engineering>
7. **Conclusion:** Summarize the main findings and conclusion (half a page).

Your project report must include all the above sections and be less than 10 pages **excluding references, followed by an appendix** containing your code and GenAI generated code. You may alternatively provide a link to your code and data on a shared repository (e.g. GitHub). Code should preferably be written in Python or R.

You must submit a **single PDF for report** and a **single pdf for presentation**.

Your presentation should include the same sections as above and be **less than 10 minutes long**.